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```
% Econ 527 Spring 2026
% HW 4
% Professor Greaney
% Written by: Camille Bergeron
% Due: May 22, 2026
```

Question 2: Hugget Model

```
run('programs/PS4_huggett.m'); % run Hugget
```

```
Transition probabilities:
```

0.7351	0.2321	0.0305	0.0021	0.0001	0.0000	0.0000
0.0387	0.7453	0.1945	0.0204	0.0011	0.0000	0.0000
0.0020	0.0778	0.7514	0.1560	0.0123	0.0004	0.0000
0.0001	0.0061	0.1170	0.7535	0.1170	0.0061	0.0001
0.0000	0.0004	0.0123	0.1560	0.7514	0.0778	0.0020
0.0000	0.0000	0.0011	0.0204	0.1945	0.7453	0.0387
0.0000	0.0000	0.0001	0.0021	0.0305	0.2321	0.7351

```
File not found, calculating equilibrium interest rate distribution...
```

Func-count	x	f(x)	Procedure
1	-0.0150246	0.00325171	initial
2	0.00659131	1.51304	golden
3	-0.028384	0.187741	golden
4	-0.0188175	0.0335984	parabolic
5	-0.0103326	0.0162749	parabolic
6	-0.0137713	0.000122134	parabolic
7	-0.0135837	1.57918e-05	parabolic
8	-0.0134659	2.48403e-07	parabolic
9	-0.0134992	5.91309e-07	parabolic
10	-0.0134326	3.12387e-06	parabolic

```
Optimization terminated:
```

```
the current x satisfies the termination criteria using OPTIONS.TolX of 1.000000e-04
```

```
Equilibrium r* = -0.013466
```

```
File not found, calculating hugget model...
```

```
Took 0.4351 seconds to converge
```

```
States at lower bound (a_min=-1.1103): 11
```

States at upper bound (a_max=55.5163): 0
File not found, calculating hugget net asset demands
Took 12.6246 seconds to calculate

PS4_run_01-eps-converted-to.pdf

PS4_run_02-eps-converted-to.pdf

PS4_run_03-eps-converted-to.pdf

PS4_run_04-eps-converted-to.pdf

Question 3: Aiyagari

```
run('programs/PS4_aiyagari.m'); % run Aiyagari
```

```
File not found | solving Aiyagari model...
iter   1 | k = 5.000000 | net_assets = 23.052835 | diff = 1.81e+01
iter   2 | k = 10.415851 | net_assets = 1.612913 | diff = 8.80e+00
iter   3 | k = 7.774969 | net_assets = 2.652661 | diff = 5.12e+00
iter   4 | k = 6.238277 | net_assets = 4.899017 | diff = 1.34e+00
iter   5 | k = 5.836499 | net_assets = 6.495597 | diff = 6.59e-01
iter   6 | k = 6.034228 | net_assets = 5.588198 | diff = 4.46e-01
iter   7 | k = 5.900419 | net_assets = 6.169092 | diff = 2.69e-01
iter   8 | k = 5.981021 | net_assets = 5.804459 | diff = 1.77e-01
iter   9 | k = 5.928052 | net_assets = 6.038617 | diff = 1.11e-01
iter  10 | k = 5.961222 | net_assets = 5.889653 | diff = 7.16e-02
iter  11 | k = 5.939751 | net_assets = 5.985155 | diff = 4.54e-02
iter  12 | k = 5.953372 | net_assets = 5.924184 | diff = 2.92e-02
iter  13 | k = 5.944616 | net_assets = 5.963225 | diff = 1.86e-02
iter  14 | k = 5.950199 | net_assets = 5.938271 | diff = 1.19e-02
iter  15 | k = 5.946620 | net_assets = 5.954240 | diff = 7.62e-03
```

```

iter    16 | k = 5.948906 | net_assets = 5.944028 | diff = 4.88e-03
iter    17 | k = 5.947443 | net_assets = 5.950561 | diff = 3.12e-03
iter    18 | k = 5.948378 | net_assets = 5.946383 | diff = 2.00e-03
iter    19 | k = 5.947780 | net_assets = 5.949056 | diff = 1.28e-03
iter    20 | k = 5.948162 | net_assets = 5.947346 | diff = 8.16e-04
iter    21 | k = 5.947917 | net_assets = 5.948440 | diff = 5.22e-04
iter    22 | k = 5.948074 | net_assets = 5.947740 | diff = 3.34e-04
iter    23 | k = 5.947974 | net_assets = 5.948188 | diff = 2.14e-04
iter    24 | k = 5.948038 | net_assets = 5.947901 | diff = 1.37e-04
iter    25 | k = 5.947997 | net_assets = 5.948084 | diff = 8.74e-05
iter    26 | k = 5.948023 | net_assets = 5.947967 | diff = 5.59e-05
iter    27 | k = 5.948006 | net_assets = 5.948042 | diff = 3.58e-05
iter    28 | k = 5.948017 | net_assets = 5.947994 | diff = 2.29e-05
iter    29 | k = 5.948010 | net_assets = 5.948025 | diff = 1.46e-05
iter    30 | k = 5.948015 | net_assets = 5.948005 | diff = 9.36e-06

```

Converged in 30 iterations (21.54 seconds)

r* = 0.022968, w* = 1.171094, K* = 5.948005

Wealth Inequality
Gini coefficient : 0.5381
Top 10% share : 0.3421 (34.2%)
Top 1% share : 0.0553 (5.5%)

States at lower bound (a_min = 0.00): 8

States at upper bound (a_max = 60.00): 3

File not found, calculating aiyagari net asset demands...

Took 16.5857 seconds to calculate

Warning: Ignoring extra legend entries.

Warning: Exported image displays axes toolbar. To remove axes toolbar from image, export again.

PS4_run_05-eps-converted-to.pdf

PS4_run_06-eps-converted-to.pdf

PS4_run_07-eps-converted-to.pdf

PS4_run_08-eps-converted-to.pdf

PS4_run_09-eps-converted-to.pdf